

Profile

My research develops **differential-geometric tools for statistics on smooth and stratified manifolds**, with a focus on **covariance matrices** and **correlation matrices**. I study how geometric choices—especially **log-Euclidean metrics** and **quotient constructions**—shape geodesics and statistical summaries (e.g., Fréchet means and regression), and how these models behave near rank-deficient limits through **orbit-type stratification**. A recurring goal is to keep the geometry mathematically faithful while remaining computationally practical for applications such as longitudinal brain connectivity.

Research interests

Riemannian and quotient geometry on smooth and stratified manifolds; log-Euclidean metrics; correlation and covariance geometry; orbit-type stratification and rank-deficient limits; geometric statistics (Fréchet means, regression, trajectories) with applications to medical imaging and neuroimaging.

Current position

Oct. 2023 – present **PhD Student in Geometric Statistics, Inria (EPIONE) & Université Côte d'Azur, France**
Thesis: *Riemannian geometry and orbit type stratification of correlation matrices.*
Advisor: Xavier Pennec.
Funding: 3IA (Interdisciplinary Institute for Artificial Intelligence).

Education

2023 **M.Sc. in Mathematical Engineering, Stochastic and Statistical Modeling**, Université Côte d'Azur, France, Highest honors
2022 **M.Sc. in Pure and Applied Mathematics, Algebra and Geometry**, Université Côte d'Azur, France
2020 **B.Sc. in Fundamental Mathematics**, Université Côte d'Azur, France

Work in preparation

In prep. **O. Bisson**, D. Loiseaux, X. Pennec. *Low-Rank Correlation Matrices and Orbit-Type Stratification of the Product of Spheres*. Manuscript in preparation.

(Pre)Publications

2025 **O. Bisson**, X. Pennec. *Log-Euclidean Lie Groups*. arXiv preprint arXiv:2511.13380.
2025 **O. Bisson**, Y. Aeschlimann, S. Deslauriers-Gauthier, X. Pennec. *Log-euclidean frameworks for smooth brain connectivity trajectories*. In: *Geometric Science of Information (GSI)*, St Malo, France, 2025, pp. 194–204. Springer. arXiv:2507.15374. *Nominated for Best Paper Award*.
2024 **O. Bisson**, X. Pennec. *Differential of the Stretch Tensor for Any Dimension with Applications to Quotient Geodesics*. *Comptes Rendus. Mathématique*, 362 (2024), pp. 1847–1856. doi:10.5802/crmath.692. PDF.

Talks & conferences

Jul. 2026 *Poster: Log-Euclidean Lie Groups and rs-fMRI Brain Connectivity Trajectories*. Foundations of Computational Mathematics (FoCM), Faculty of Mathematics, University of Vienna, Vienna, Austria, 8–18 July 2026.
May 2026 *Log-Euclidean Geometry of Full-Rank Correlation Matrices*. Geometry, Topology and Machine Learning 2026 (GML 2026), Sorbonne Université, Paris, France, 18–22 May 2026.

- Dec. 2025 *Mini-course: Quotient Spaces and Related Differential Geometry*. EPIONE Seminar, Inria Sophia Antipolis, France.
- Oct. 2025 *Log-Euclidean Frameworks for Smooth Brain Connectivity Trajectories*. Geometric Science of Information (GSI), St Malo, France.
- Mar. 2025 *Geometry, Stratification and Applications of Structured Correlation Matrices*. EPIONE Seminar, Inria Sophia Antipolis, France.
- Feb. 2025 *Differentiable Structures on Correlation Matrices and Applications in Neuroimaging. Infinite-dimensional Geometry: Theory and Applications*, ESI (Vienna), Austria.
- Dec. 2024 *Geometry, Stratification and Applications of Structured Correlation Matrices*. 3IA Côte d'Azur PhD Seminar, Nice, France.
- Nov. 2024 *Riemannian Metrics on Correlation Matrices with Applications in Neuroimaging*. Mathematical Imaging and Random Geometry (RT MAIAGES), Nice, France.
- Oct. 2024 *Introduction to the Riemannian Geometry of Correlation Matrices*. EPIONE Seminar, Inria Sophia Antipolis, France.
- Feb. 2024 *A framework to put metrics on quotient spaces, with application in medical image analysis*. EPIONE Annual Retreat, Auron, France.
- Dec. 2023 *Poster: Geometry, stratification and applications of structured correlation matrices*. 3IA Scientific Day, SKEMA, Sophia Antipolis, France.
- Oct. 2023 *A framework to put metrics on quotient spaces, with application in medical image analysis*. EPIONE Seminar, Inria Sophia Antipolis, France.

Awards & Funding

- 2023–2026 **3IA Côte d'Azur PhD Funding**, Université Côte d'Azur / Inria.
- 2025 **Best Paper Award nomination** (GSI 2025) for *Log-euclidean frameworks for smooth brain connectivity trajectories*.

Teaching

- 2025–2026 **Teaching assistant (TD)**, *Introduction à l'analyse (L1)*, Université Côte d'Azur
Sets and functions; limits and continuity; differentiability; function study; injective/surjective/bijective maps; integrals and primitives.
- 2024–2025 **Teaching assistant (TD)**, *Compléments d'algèbre linéaire (L2)*, Université Côte d'Azur
Orthogonality and projections; Gram–Schmidt; least squares; isometries; introduction to PCA.
- 2024–2025 **Teaching assistant (TD)**, *Calculus II (L2)*, Université Côte d'Azur
Complex numbers; linear ODEs; multivariable calculus (partials, gradient, tangent planes, level sets, local extrema).
- 2023–2024 **Teaching assistant (TD)**, *Maths Approfondissement 2 (L1)*, Université Côte d'Azur
Exponential/logarithm refreshers; linear ODEs; Riemann integration basics; partial fractions; Euclidean vector geometry.

Software & open-source

Contributor to the `geomstats` Python package, with implementations related to Riemannian geometry on SPD and correlation matrix manifolds.

Previous Research experiences

- Apr. 2023 – Oct. 2023 **Research intern**, Inria (EPIONE), Université Côte d'Azur, Nice/Valbonne, France
Topic: *Interpolation and extrapolation of correlation matrices for dynamic functional connectivity*.
Supervisors: Xavier Pennec, Samuel Deslauriers-Gauthier.
Implementation and evaluation of Riemannian metrics on correlation matrices for geodesic interpolation and extrapolation on resting-state fMRI correlation time-series.

- Sep. 2022 – Feb. 2023 **Research project**, LJAD & CEPAM, Université Côte d'Azur, Nice, France
Topic: *Supervised classification of wood charcoal fragments: from machine to active learning.*
Supervisor: Marco Corneli.
 Development of learning methods to automatically identify wood charcoal remains from a Southern-African context (M.Sc. capstone project).
- Mar. 2022 – Jun. 2022 **Research intern**, LJAD, Université Côte d'Azur, Nice, France
Topic: *Quantum representations of mapping class groups.*
Supervisor: Jeremy Toulisse.
 Topological/combinatorial approach (skein theory) to construct quantum representations of mapping class groups.

Languages

French	Native	<i>Fluent</i>
English	Fluent	<i>Reading, writing, speaking</i>
Italian	Basic	<i>Working knowledge</i>

Additional

Outside research: Volleyball, rock climbing, sailing, beach volleyball and swimming. I am deeply convinced that **anyone** can do mathematics.